



Clusterization

KMeans algorithm

Introduction

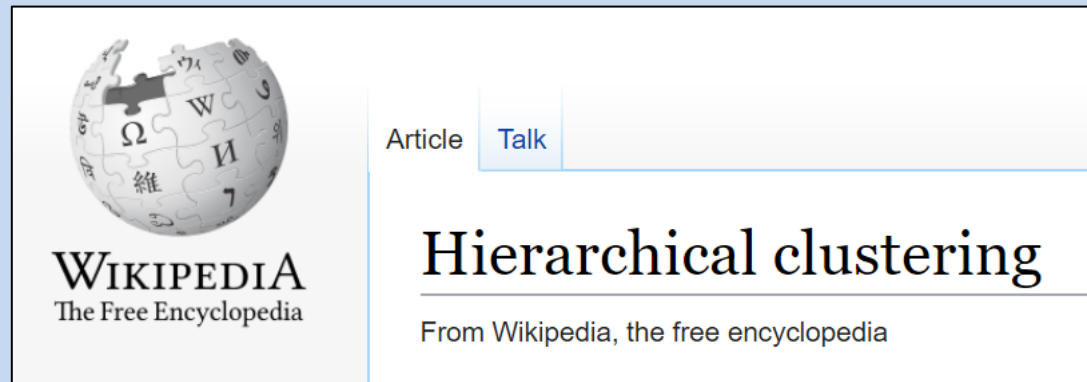


■ *KMeans* algorithm:



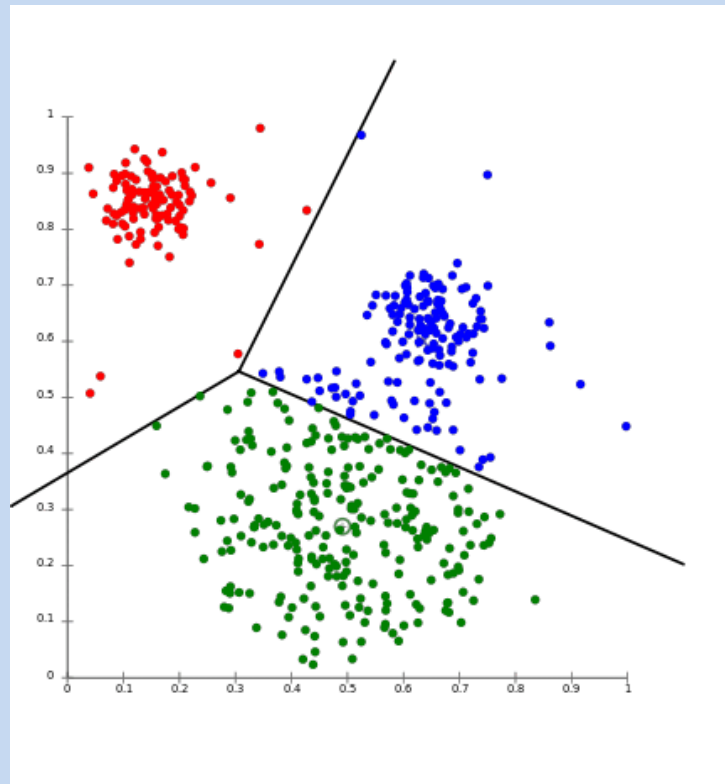
■ *Hierarchical clustering*:

- *agglomerative*,
- *divisive*





KMeans

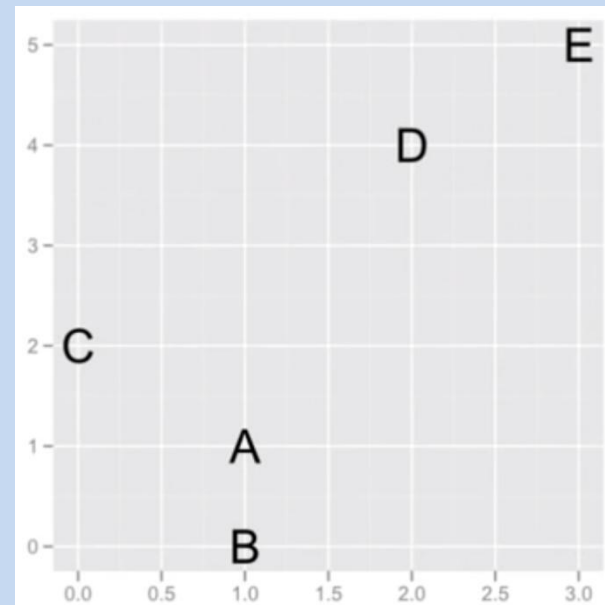


KMeans



■ Input data:

i	X_1	X_2
A	1	1
B	1	0
C	0	2
D	2	4
E	3	5



■ Pre-conditions:

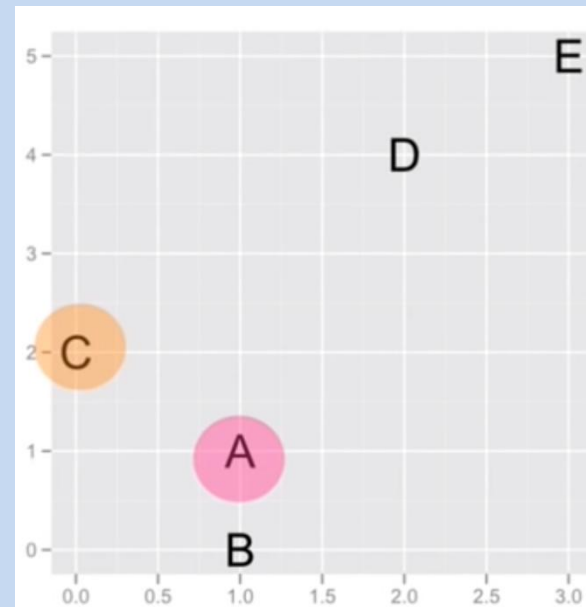
- two clusters ($k=2$),
- objects A and C are randomly selected as the initial means.

KMeans



■ Input data:

$\bar{X}_1^0$	i	X_1	X_2
	A	1	1
	B	1	0
	C	0	2
	D	2	4
	E	3	5
$\bar{X}_2^0$			
			



■ Pre-conditions:

- two clusters ($k=2$),
- objects **A** and **C** are randomly selected as the **initial means**.

KMeans



- Computing the distances* between:
 - (a) each of the cluster **means**, and
 - (b) all other **points (objects)**:

	i	X_1	X_2
$\bar{X}_1^0$ 	A	1	1
	B	1	0
$\bar{X}_2^0$ 	C	0	2
	D	2	4
	E	3	5

i		
A		
B		
C		
D		
E		

* **Euclidean** distance

KMeans



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 - (a) each of the cluster **means**, and
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i		
A	0	
B		
C		0
D		
E		

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KMeans



- Computing the distances* between:
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	E	3	5

i	1	2
A	0	1.4
B	1	2.2
C	1.4	0
D	3.2	2.8
E	4.5	4.2

Distances between objects A to E and the centroid of the 2nd cluster

* **Euclidean distance**

KMeans



- Computing the distances* between:
 - (a) each of the cluster **means**, and
 - (b) all other **points** (objects):

	i	A <-> A	
$\bar{X}_1^0$ 	A	1	1
	B	1	0
$\bar{X}_2^0$ 	C	0	2
	D	2	4
	E	3	5

i	1	2
A	0	1.4
B	1	2.2
C	1.4	0
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E	4.5	4.2

* **Euclidean** distance

KMeans



- Computing the distances* between:
 - (a) each of the cluster **means**, and
 - (b) all other **points** (objects):

	i	A <-> B	
$\bar{X}_1^0$ 	A	1	1
	B	1	0
$\bar{X}_2^0$ 	C	0	2
	D	2	4
	E	3	5

i	1 	2 
A	0	1.4
B	1	2.2
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* **Euclidean** distance

KMeans



- Computing the distances* between:
 - (a) each of the cluster **means**, and
 - (b) all other **points (objects)**:

	i	X_1	X_2
$\bar{X}_1^0$ 	A	B <-> C	
	B	1	0
$\bar{X}_2^0$ 	C	0	2
	D	2	4
	E	3	5

i	1	2
A	0	1.4
B	1	2.2
C	1.4	0
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KMeans



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$\bar{X}_2^0$ 	C	0	2
	D	2	4
	E	3	5

	1	2	
A	0	1.4	1
B	1	2.2	1
C	1.4	0	2
D	3.2	2.8	2
E	4.5	4.2	2

* **Euclidean** distance

KMeans



- Assigning each case to the cluster with the **closest mean**
- Recalculation of all cluster **means**:

i	1	2	Cluster
A	0	1.4	
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KMeans



- Assigning each case to the cluster with the **closest mean**
- Recalculation of all cluster **means**:

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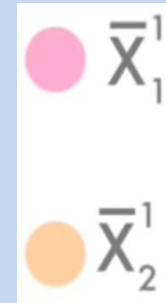
KMeans



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KMeans



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 $\bar{X}_1^1$

 $\bar{X}_2^1$

KMeans



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i	X_1	X_2
A	1	1
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● $\bar{X}_1^1 = (1, 0.5)$

● $\bar{X}_2^1$

KMeans



- Assigning each case to the cluster with the **closest mean**
- Recalculation of all cluster **means**:

i	1	2	Cluster
A	0	1.4	1
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E	4.5	4.2	2

i	X_1	X_2
A	1	1
B	1	0
C	0	2
D	2	4
E	3	5

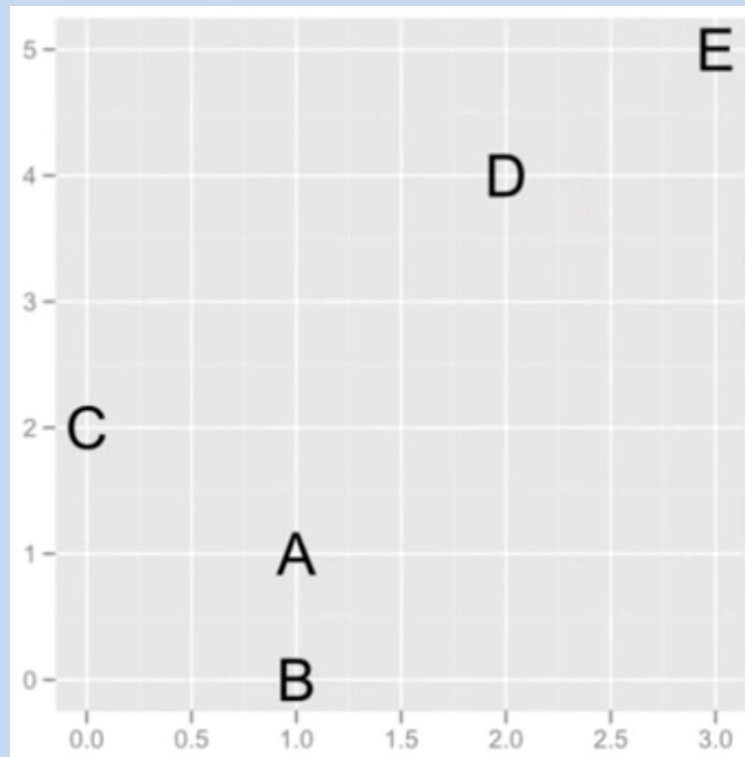
● $\bar{X}_1^1 = (1, 0.5)$

● $\bar{X}_2^1 = (1.7, 3.7)$

KMeans



■ Visualisation:



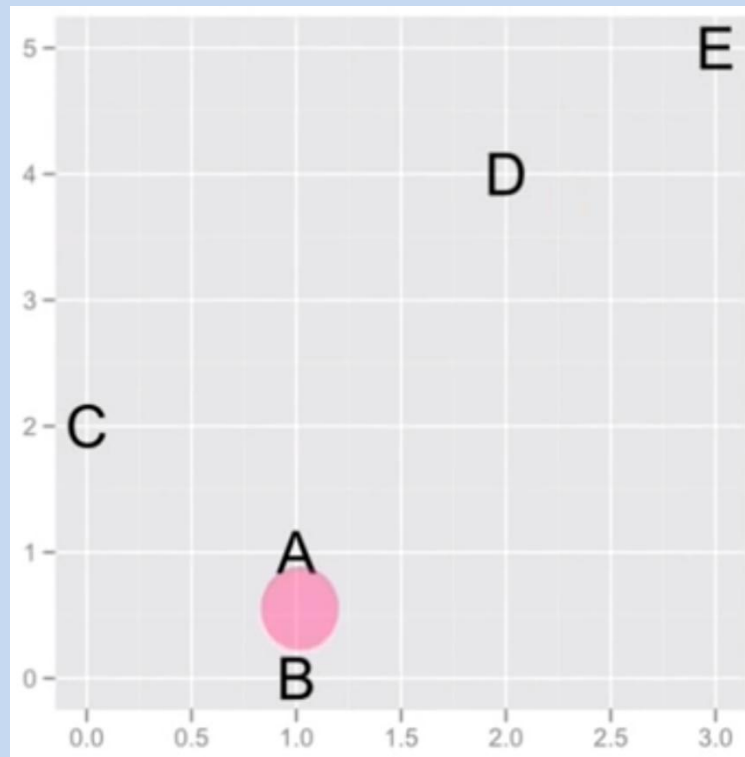
● $\bar{X}_1^1 = (1, 0.5)$

● $\bar{X}_2^1 = (1.7, 3.7)$

KMeans



■ Visualisation:



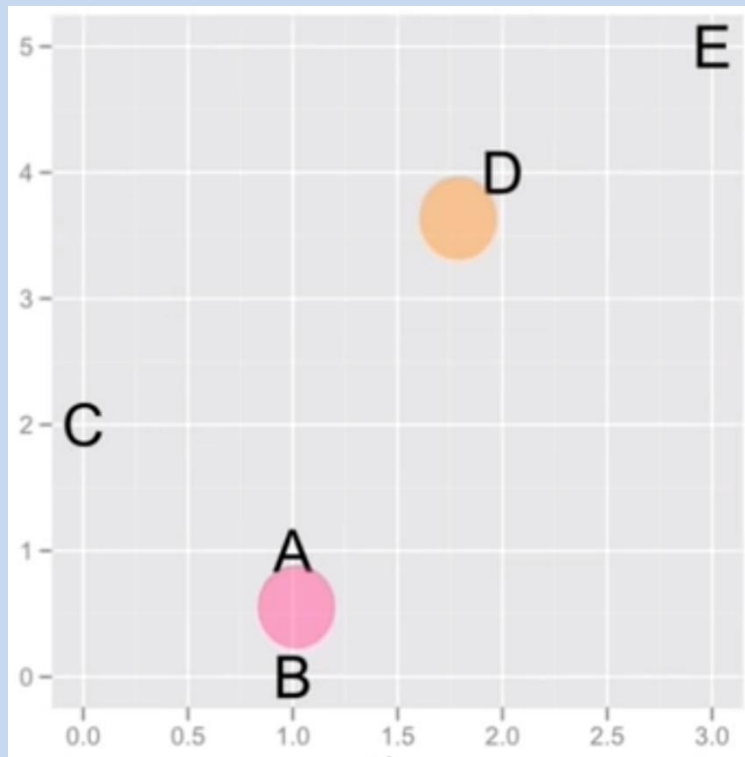
● $\bar{X}_1^1 = (1, 0.5)$

● $\bar{X}_2^1 = (1.7, 3.7)$

KMeans



■ Visualisation:



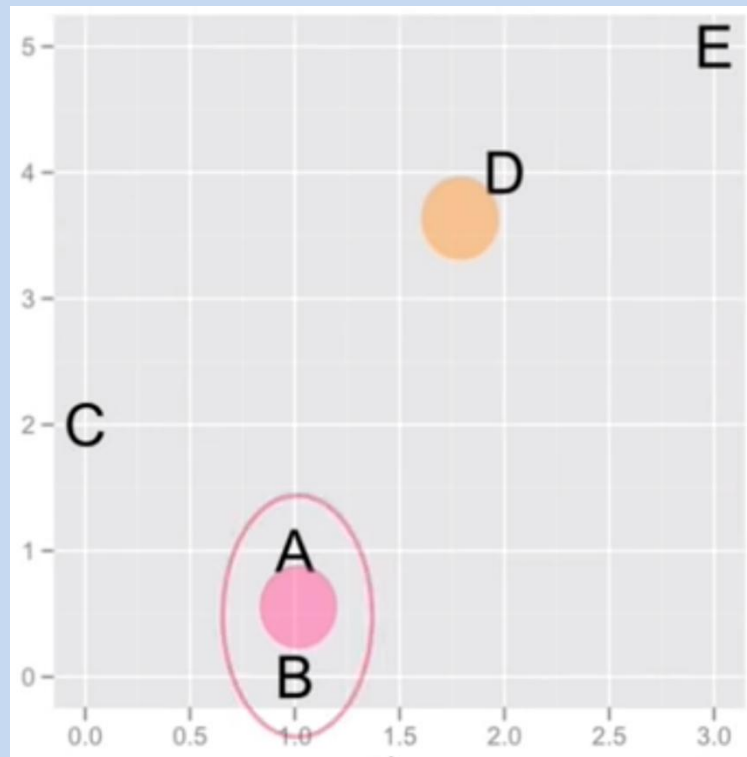
● $\bar{X}_1^1 = (1, 0.5)$

● $\bar{X}_2^1 = (1.7, 3.7)$

KMeans



■ Visualisation:



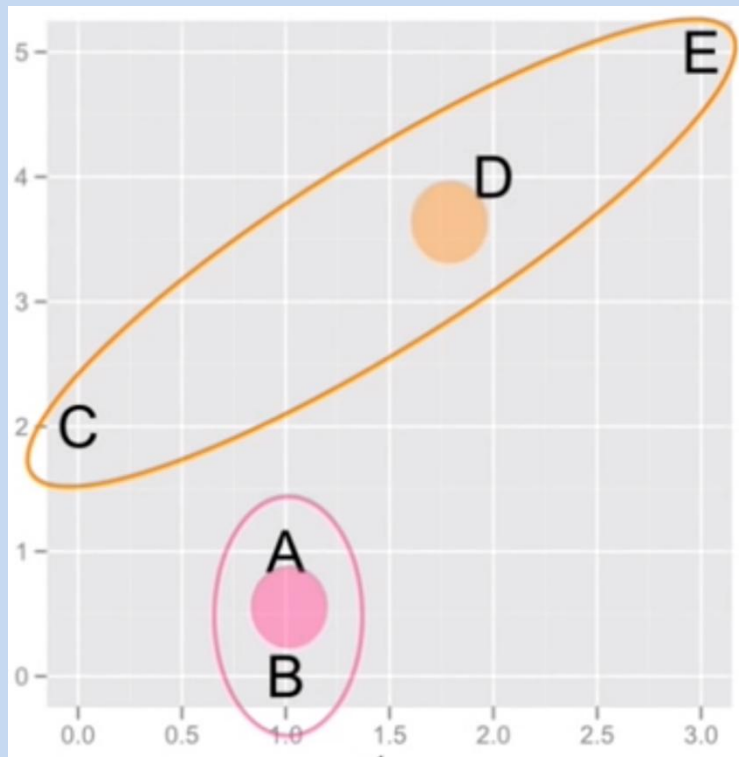
● $\bar{X}_1^1 = (1, 0.5)$

● $\bar{X}_2^1 = (1.7, 3.7)$

KMeans



■ Visualisation:



● $\bar{X}_1^1 = (1, 0.5)$

● $\bar{X}_2^1 = (1.7, 3.7)$

KMeans



- Computing the distances* between:
 - (a) each of the cluster **means**, and
 - (b) all other **points** (**objects**):

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● $\bar{X}_2^1 = (1.7, 3.7)$

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A		
B		
C		
D		
E		

* **Euclidean** distance

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 - (a) each of the cluster **means**, and
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E	3	5

● $\bar{X}_1^1 = (1, 0.5)$

● $\bar{X}_2^1 = (1.7, 3.7)$

i	1	2
A	0.5	2.7
B	0.5	3.7
C	1.8	2.4
D	3.6	0.5
E	4.9	1.9

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 - (a) each of the cluster **means**, and
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Distances between objects A to E and the centroid of the 2nd cluster

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* **Euclidean** distance

KMeans



- Assigning each case to the cluster with the **closest mean**
- Recalculation of all cluster **means**:

i	1	2	Cluster
A	0.5	2.7	1
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C	1.8	2.4	1
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i	X_1	X_2
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B	1	0
C	0	2
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E	3	5

● $\bar{X}_1^1$

● $\bar{X}_2^1$

KMeans



- Assigning each case to the cluster with the **closest mean**
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A	0.5	2.7	1
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KMeans



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E	4.9	1.9	2

i	X_1	X_2
A	1	1
B	1	0
C	0	2
D	2	4
E	3	5

● $\bar{X}_1^2 = (0.7, 1)$

● $\bar{X}_2^2$

KMeans



- Assigning each case to the cluster with the **closest mean**
- Recalculation of all cluster **means**:

i	1	2	Cluster
A	0.5	2.7	1
B	0.5	3.7	1
C	1.8	2.4	1
D	3.6	0.5	2
E	4.9	1.9	2

i	X_1	X_2
A	1	1
B	1	0
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D	2	4
E	3	5

● $\bar{X}_1^2 = (0.7, 1)$

● $\bar{X}_2^2$

KMeans



- Assigning each case to the cluster with the **closest mean**
- Recalculation of all cluster **means**:

i	1	2	Cluster
A	0.5	2.7	1
B	0.5	3.7	1
C	1.8	2.4	1
D	3.6	0.5	2
E	4.9	1.9	2

i	X ₁	X ₂
A	1	1
B	1	0
C	0	2
D	2	4
E	3	5

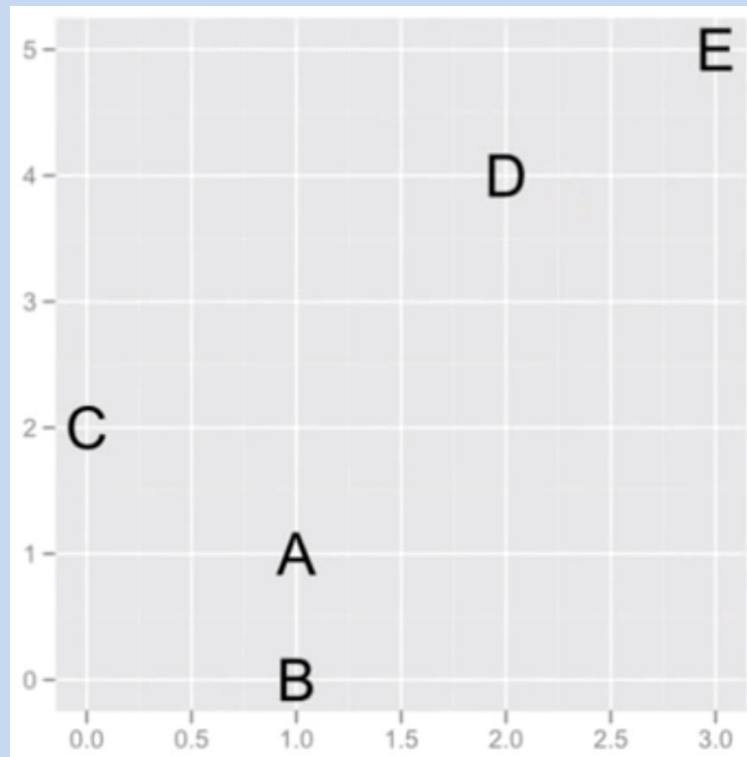
● $\bar{X}_1^2 = (0.7, 1)$

● $\bar{X}_2^2 = (2.5, 4.5)$

KMeans



■ Visualisation:



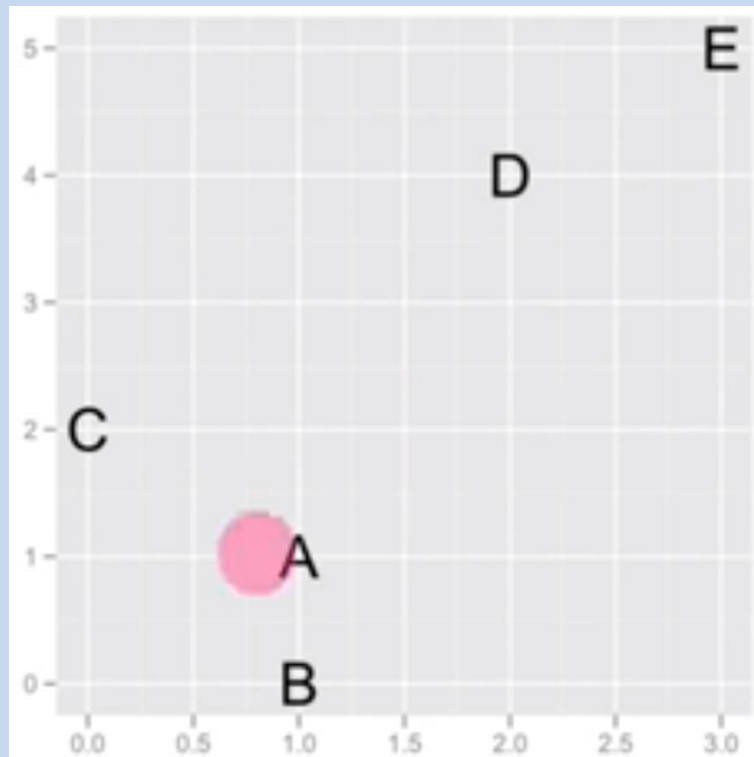
● $\bar{X}_1^2 = (0.7, 1)$

● $\bar{X}_2^2 = (2.5, 4.5)$

KMeans



■ Visualisation:



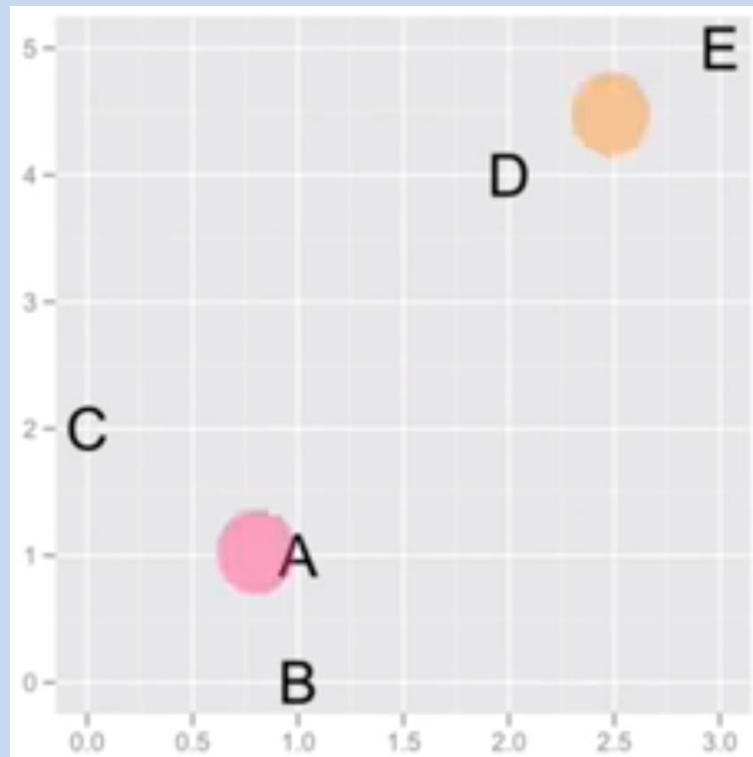
● $\bar{X}_1^2 = (0.7, 1)$

● $\bar{X}_2^2 = (2.5, 4.5)$

KMeans



■ Visualisation:



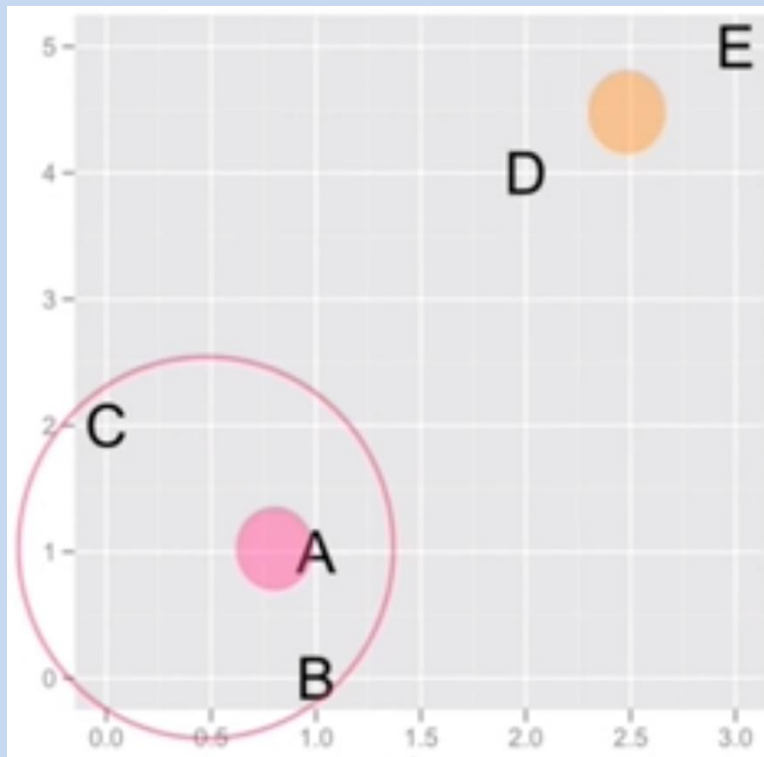
● $\bar{X}_1^2 = (0.7, 1)$

● $\bar{X}_2^2 = (2.5, 4.5)$

KMeans



■ Visualisation:



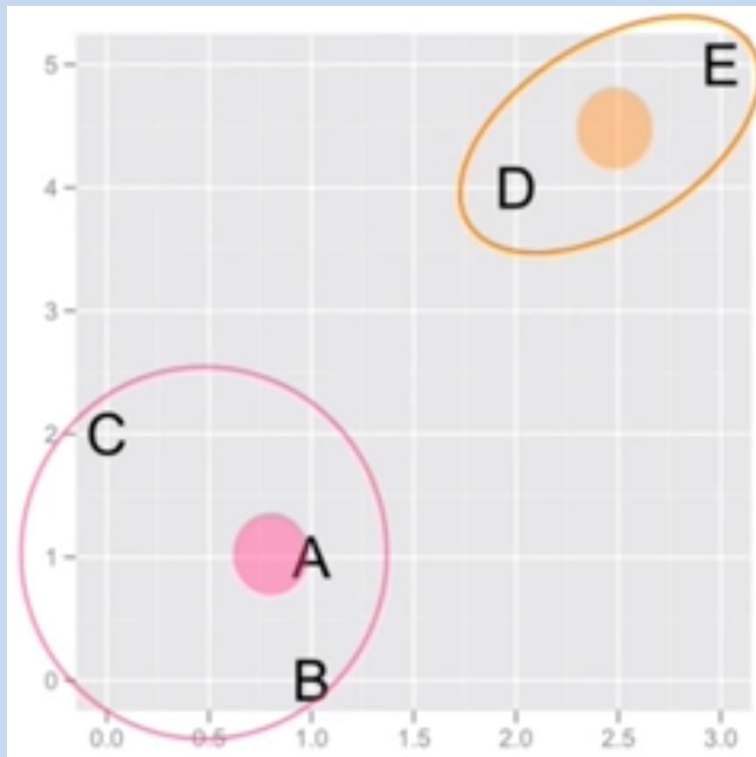
● $\bar{X}_1^2 = (0.7, 1)$

● $\bar{X}_2^2 = (2.5, 4.5)$

KMeans



■ Visualisation:



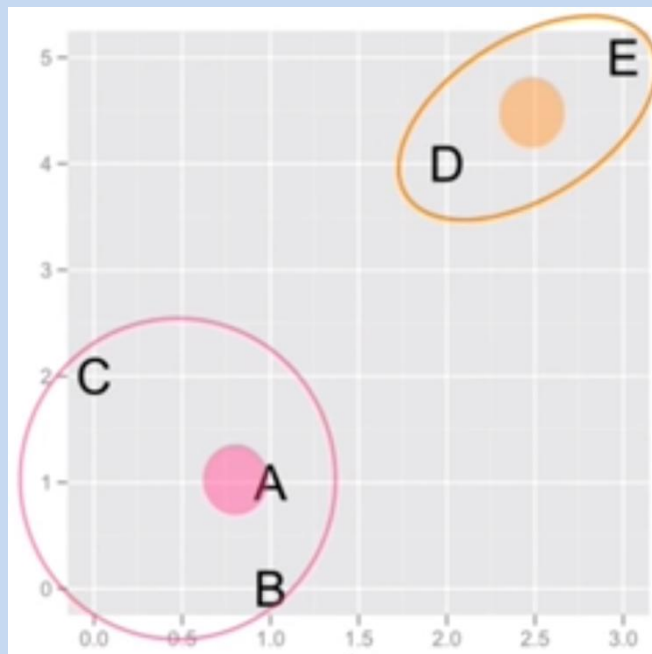
● $\bar{X}_1^2 = (0.7, 1)$

● $\bar{X}_2^2 = (2.5, 4.5)$

KMeans



- Assigning each case to the cluster with the **closest mean**
- Recalculation of all cluster **means**
- The algorithm has converged – i.e., re-calculating the distances, re-assigning of cases to clusters results does not change. Thus, that is the **final solution**:





*Thank you for your
attention...*

